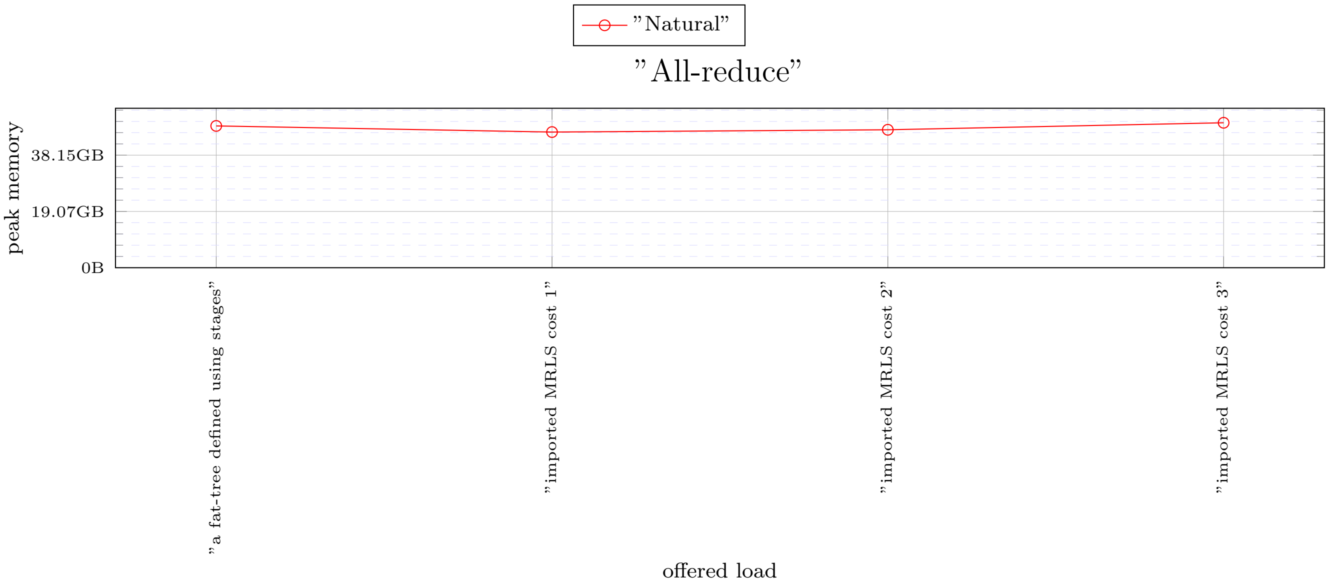

Figure 1: X: "All-reduce"

The following versions used in the simulations.

- heads/alex-dev-8e2a39f34b812569e780c03ed6a7e158aa4b09bc(0.6.3)
- heads/alex-dev-a8bf2d5ee67d9d0c7dceda54f01ab7cede4ab269(0.6.3)

```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 9999999999999,
  measured: 9999999999999,
  general_frequency_divisor: 2,
  topology: topo![
    MultiStage{
      stages: [
        Fat{ bottom_factor: 18, top_factor: 18 },
        Fat{ bottom_factor: 18, top_factor: 18 },
        Fat{ bottom_factor: 18, top_factor: 18 }],
      servers_per_leaf: 18,
      legend_name: "a fat-tree defined using stages"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_5832_18_2916_36" }],
      servers_per_leaf: 18,
      legend_name: "imported MRLS cost 1"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_8748_24_5832_36" }],
      servers_per_leaf: 12,
      legend_name: "imported MRLS cost 2"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_11664_27_8748_36" }],
      servers_per_leaf: 9,
      legend_name: "imported MRLS cost 3"}],
  traffic: ![
    TrafficMap{
      tasks: 104976,
      map: ![
        Composition{
          patterns: [
            Identity{ legend_name: "Natural" },
            CartesianEmbedding{
              source_sides: [ 65536 ],
              destination_sides: [ 104976 ]}],
          middle_sizes: [ 65536, 104976 ],
          legend_name: "Natural"}],
      application: ![
        AllReduce{ tasks: 65536, data_size: 65536 }],
      legend_name: "All-reduce"}],
  maximum_packet_size: 16,
  router: InputOutput{
    virtual_channels: topo![ 3, 4, 4, 4 ],
    virtual_channel_policies: topo![
      [
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
        LowestLabel,
        EnforceFlowControl,
        Random],
      [
        MapHop{
          hop_to_policy: [
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 3 ]},
            ArgumentVC{
              allowed: [ 3 ]}]},
        VecLabel{
          label_vector: [ 0, 64, 80 ]},
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
        LowestLabel,
        EnforceFlowControl,
        Random],
      [
        MapHop{
          hop_to_policy: [
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              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 2 ]},
```

```
          ArgumentVC{
            allowed: [ 3 ]},
          ArgumentVC{
            allowed: [ 3 ]}]},
        VecLabel{
          label_vector: [ 0, 64, 80 ]},
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
        LowestLabel,
        EnforceFlowControl,
        Random],
      [
        MapHop{
          hop_to_policy: [
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            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 3 ]},
            ArgumentVC{
              allowed: [ 3 ]}]},
        VecLabel{
          label_vector: [ 0, 64, 80 ]},
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
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        EnforceFlowControl,
        Random]],
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    buffer_size: 128,
    bubble: false,
    flit_size: 16,
    intransit_priority: false,
    allow_request_busy_port: true,
    output_buffer_size: 64,
    crossbar_frequency_divisor: 1,
    crossbar_delay: 2},
  routing: topo![
    UpDown{ legend_name: "minimal routing" },
    Polarized{ include_labels: true, legend_name: "Polarized original" },
    Polarized{ include_labels: true, legend_name: "Polarized original" },
    Polarized{ include_labels: true, legend_name: "Polarized original" }],
  link_classes: [
    LinkClass{ delay: 2 },
    LinkClass{ delay: 2 },
    LinkClass{ delay: 2 },
    LinkClass{ delay: 2 }],
  launch_configurations: [
    Slurm{ job_pack_size: 1, time: "35-00:00:00", mem: "70G" }]}
```